TechPost ELEVATED

DC – AI – Routing – Switching – Security – Wireless – Automation

ACX 7020 Routers Deepdive

An HPE Juniper Networking Techpost

Dominique Blanchard

2026-03-16

HPE

Contents

1 ACX 7020 Routers Deepdive 2
1.1 High-Level Introduction . . . 2
1.2 Naming convention . . . 2
1.3 SKUs . . . 3
1.4 Environment Standards . . . 4
1.5 ACX7020 Hardware Overview . . . 4
1.6 Software . . . 4
1.7 Oversubscription . . . 5
1.8 ACX7020 Ethernet Ports . . . 5
1.9 Impact of Timing Activation . . . 6
1.10 Port Checker Tool . . . 6
1.11 Supported Port FEC . . . 6
1.12 Serdes Rate . . . 7
1.13 Supported Optics . . . 8
1.14 Internal Description: Block Diagrams . . . 8
1.15 No-FRU System and Redundancy . . . 9
1.16 Routing Engine . . . 9
1.17 Fans . . . 10
1.18 AC and DC Power . . . 10
1.19 Timing Support . . . 11
1.20 MDB Profile support . . . 11
1.21 Conclusion . . . 12
1.22 Useful links . . . 12
1.23 Glossary . . . 13
1.24 Acknowledgments . . . 13

1 ACX 7020 Routers Deepdive

Dominique Blanchard - 03/16/2026

1.1 High-Level Introduction

The ACX7020 is a HPE Juniper Networking product powered by Broadcom Qumran Q2N [BCM88295] and running Junos EVO modular software architecture. It offers a mix of 16 ports 1/10GbE and 4 ports 1/10/25GbE.



This 1RU Ethernet-only router offers 100Gbps bidirectional forwarding capacity and with Class-C Timing support.

It addresses primarily the Cable DAA/UAA use cases, Mobile xHaul Transport market, Converged Metro but can be used in multiple roles as soon as features and scales are meeting the requirements.

ACX7020 is a temperature hardened box supporting (operating) temperature from -40C to +65C and from 5 to 90% non-condensing (at an altitude of 6000 Feet)

Port Speed	Maximum Number of Ports on ACX7020(Details / Total)
25GbE	4
10GbE	16+4
1GbE	16+4

*Maximum number of ports supported on ACX7**020** Routers*

1.2 Naming convention

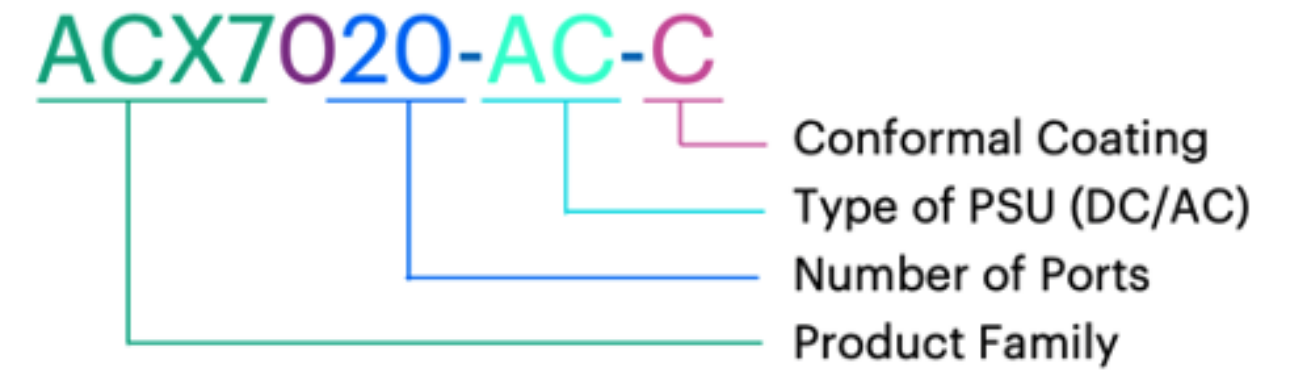
We have a total of 4 SKUs (Stock Keeping Units) with the following characteristics:

- Ports density: 4x 1/10/25GbE and 16x 1/10GbE.

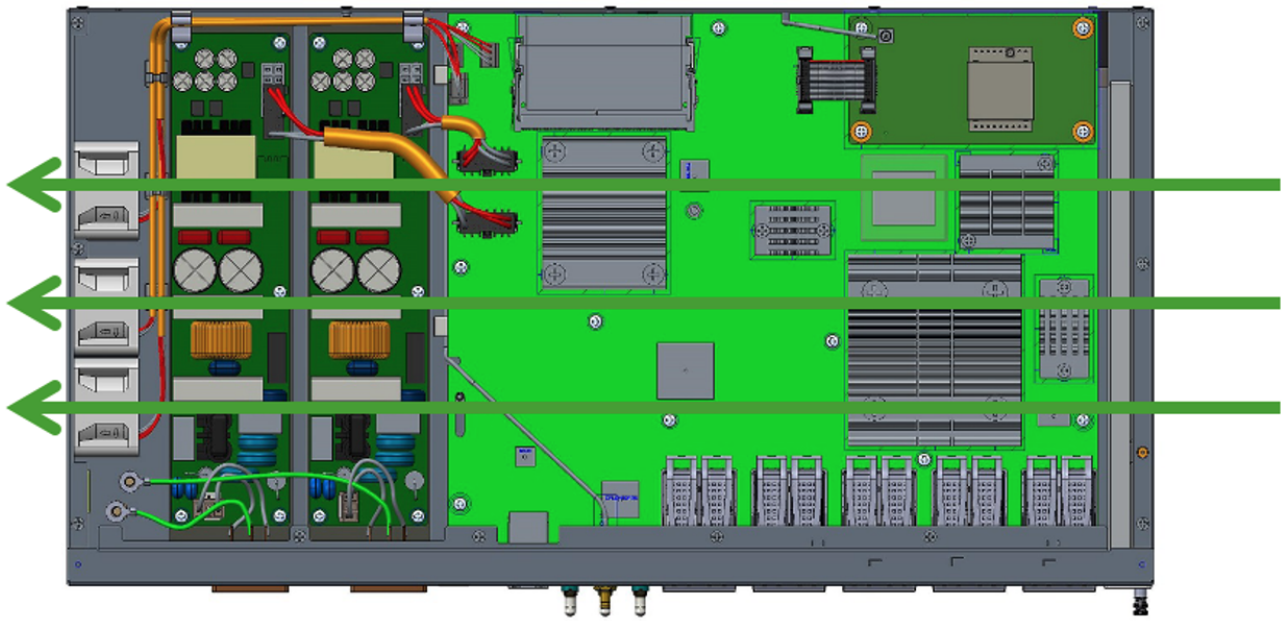
Port Number 0 to 3 [SFP28 1/10/25GbE]

Port Number 4 to 19 [SFP+ 1/10GbE] - Power: 1+1 AC or DC (PSU) -24VDC to -60V VDC, Typical power 60W - Airflow: S2S (Air Flow Side to Side)

Based on these parameters, the naming logic for the product itself and the SKU is the following.



The ACX7020 is only available in Side-to-Side version as illustrated in the figure 3 below.



1.3 SKUs

ACX7020 is available with two fixed PSU, AC or DC and with, or without conformal coating. The combinations give us a total of 4 SKUs:

SKU	Description
ACX7020-AC	ACX7020 with PTP (SyncE+Transperent Clock+Boundary Clock),16GB DRAM, redundant AC PSU
ACX7020-DC	ACX7020 with PTP (SyncE+Transperent Clock+Boundary Clock),16GB DRAM, redundant DC PSU feed
ACX7020-AC-C	ACX7020 with PTP (SyncE+Transperent Clock+Boundary Clock),16GB DRAM, redundant AC PSU, Conformal coating

SKU	Description
ACX7020-DC-C	ACX7020 with PTP (SyncE+Transperent Clock+Boundary Clock),16GB DRAM, redundant DC PSU feed, Conformal coating

A few additional/optional SKUs:

Optional SKU	Description
ACX7020-FLTR	ACX7020 Air Filter, Spare
ACX702X-RMK-21	ACX7020 / ACX7024 ETSI 21inch Rack Mount Kit, Spare
ACX702X-RMK-23	ACX7020 / ACX7024 23inch Rack Mount Kit, Spare

1.4 Environment Standards

ACX7020 is the temperature hardened router (-40C to 70C at 1,000 ft) and is compliant to the following environmental standards

- IEEE1613:2013 Smart Grid – Power Substations
- IEC 61850-3:2013 Telecom equipment in Power station and substation environments
- GR 63 issue 5
- GR 3108- Class 2 Edition 3 (65C at 6,000 ft)
- GR 3108- Class 2 Edition 3 (70C at 1,000 ft)

We invite you to take a look at the TechLibrary hardware guide for more details: https://www.juniper.net/documentation/us/en/hardware/acx7020/index.html

1.5 ACX7020 Hardware Overview

```
regress@rtme-acx7020-01> show chassis hardware
Hardware inventory:
Item Version Part number Serial number Description
Chassis HU3525AX0057 JNP7020_AC [ACX7020_AC]
PSM 0 BUILTIN BUILTIN JPSU-220W-AC
PSM 1 BUILTIN BUILTIN JPSU-220W-AC
Routing Engine 0 REV 11 650-179212 HU3525AX0057 RE-ACX-7020
CB 0 BUILTIN BUILTIN ACX7020 Control Board
FPC 0 BUILTIN BUILTIN ACX7020-FPC
  PIC 0 BUILTIN BUILTIN MRATE- 4xSFP28 + 16xSFP+
Fan Tray 0 BUILTIN BUILTIN ACX7020 Fan, Right to Left Airflow - AFO
```

1.6 Software

The minimum Junos release required to operate ACX7020 is 24.4R2-EVO.

Image is named as: junos-evo-install-acx-f-x86-64–EVO

```
regress@rtme-acx7020-01> show version
Hostname: rtme-acx7020-01
Model: ACX7020_AC
Junos: 25.2R1.8-EVO
Yocto: 4.0.22
Linux Kernel: 5.15.164-10.22.33.18-yocto-standard-juniper-16986-g445edc512bb4
JUNOS-EVO OS 64-bit [junos-evo-install-acx-f-x86-64-25.2R1.8-EVO]
```

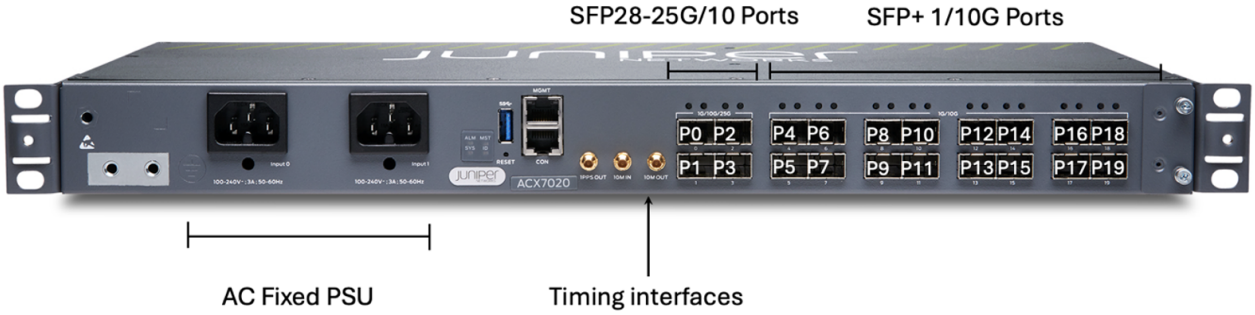
Both ACX7020 using “acx-f” images (f for fixed). “acx-m” will be used for modular systems.

1.7 Oversubscription

ACX7020 is an oversubscribed router. Total revenue port speed available ports is 260Gbps (250 is PTP in use). Broadcom Q2N chip interface bandwidth limit is 100Gbps.

1.8 ACX7020 Ethernet Ports

ACX7020 Front Panel view and Port Numbering



Port Number	Optic Type	Supported Port Speed	Default Port Speed
0 to 3	SFP28	4x25,4x10G,4x1Gbbps	25Gbps
4 to 19	SFP/SFP+	16x10G,4x1Gbps	10Gbps

Check the port capabilities with CLI “show chassis pic fpc-slot 0 pic-slot 0”

```
regress@rtme-acx7020-01> show chassis pic fpc-slot 0 pic-slot 0
FPC slot 0, PIC slot 0 information:
  Type MRATE- 4xSFP28 + 16xSFP+
  State Online
  PIC version 00.00
  Uptime 3 hours, 33 seconds

Port speed information:

  Port PFE Capable Port Speeds
  0 0 1x25G 1x10G 1x1G
  1 0 1x25G 1x10G 1x1G
  2 0 1x25G 1x10G 1x1G
  3 0 1x25G 1x10G 1x1G
  4 0 1x10G 1x1G
  5 0 1x10G 1x1G
  6 0 1x10G 1x1G
  7 0 1x10G 1x1G
  8 0 1x10G 1x1G
  9 0 1x10G 1x1G
  10 0 1x10G 1x1G
  11 0 1x10G 1x1G
  12 0 1x10G 1x1G
```

```
13 0 1x10G 1x1G
14 0 1x10G 1x1G
15 0 1x10G 1x1G
16 0 1x10G 1x1G
17 0 1x10G 1x1G
18 0 1x10G 1x1G
19 0 1x10G 1x1G
20 NA 1x10G
```

1.9 Impact of Timing Activation

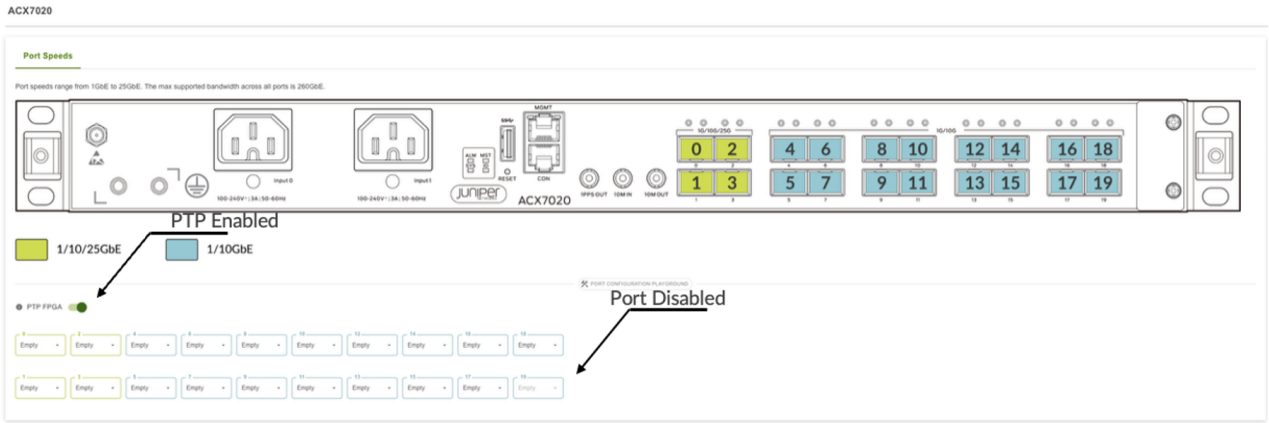
Once PTP is enabled with below command, port 19 will not be available to use.

```
regress@rtme-acx7020-01# set chassis fpc 0 pic 0 ptp-mode ## PTP mode enabled

[edit]
regress@rtme-acx7020-01# commit
commit complete

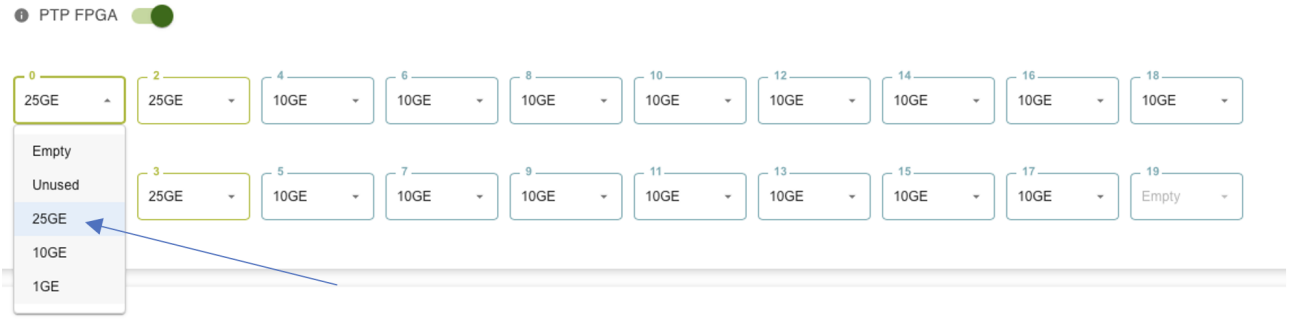
[edit]
regress@rtme-acx7020-01# run show interfaces terse et-0/0/19
error: device et-0/0/19 not found
```

Once PTP config is committed, HW FPGA on port et-0/0/19 will be used in PTP circuitry.



1.10 Port Checker Tool

The port checker (https://apps.juniper.net/port-checker/acx7020/) is a very handy tool allowing you to test the different port speed combination and identify the potential restrictions like PTP/Oversubscription.



1.11 Supported Port FEC

Port Speed	Supported FEC
25GE	FEC74, FEC91, none
10GE	FEC74, FEC91, none

1.12 Serdes Rate

SerDes or serializer/deserializer is an integrated circuit (IC or chip) transceiver that converts parallel data to serial data and vice-versa. In the chart below, we display the SerDes rate for respective port speed.

Keep in mind that port speed and SerDes rates are different, the second category is Broadcom internal rates: 10.3125Gbps, 25.7825Gbps, 1.25Gbps…

SFP Ports 0 to 19	Q2 SerDes (Gbps)
25GbE	25.78125 (x1)
10GbE	10.3125 (x1)
1GbE	1.25 (x1)

```
[vrf:none] regress@rtme-acx7020-01:~# cli-pfe
regress@rtme-acx7020-01:pfe> show picd config

pic_info_table :
           default config config config computed
pic_or_port speed pic_mode port_speed valid speed supported_speeds hidden_speeds
----------- ------- -------- ---------- ------ -------- -------------------- -------------
pic-0/0 - - - yes -
port-0/0/0 1x10G - - - 1x10G [ 1x25G 1x10G 1x1G ] [ ]
port-0/0/1 1x10G - - - 1x10G [ 1x25G 1x10G 1x1G ] [ ]
port-0/0/2 1x10G - - - 1x10G [ 1x25G 1x10G 1x1G ] [ ]
port-0/0/3 1x10G - - - 1x10G [ 1x25G 1x10G 1x1G ] [ ]
port-0/0/4 1x10G - - - 1x10G [ 1x10G 1x1G ] [ ]
port-0/0/5 1x10G - - - 1x10G [ 1x10G 1x1G ] [ ]
port-0/0/6 1x10G - - - 1x10G [ 1x10G 1x1G ] [ ]
port-0/0/7 1x10G - - - 1x10G [ 1x10G 1x1G ] [ ]
port-0/0/8 1x10G - - - 1x10G [ 1x10G 1x1G ] [ ]
port-0/0/9 1x10G - - - 1x10G [ 1x10G 1x1G ] [ ]
port-0/0/10 1x10G - - - 1x10G [ 1x10G 1x1G ] [ ]
port-0/0/11 1x10G - - - 1x10G [ 1x10G 1x1G ] [ ]
port-0/0/12 1x10G - - - 1x10G [ 1x10G 1x1G ] [ ]
port-0/0/13 1x10G - - - 1x10G [ 1x10G 1x1G ] [ ]
port-0/0/14 1x10G - - - 1x10G [ 1x10G 1x1G ] [ ]
port-0/0/15 1x10G - - - 1x10G [ 1x10G 1x1G ] [ ]
port-0/0/16 1x10G - - - 1x10G [ 1x10G 1x1G ] [ ]
port-0/0/17 1x10G - - - 1x10G [ 1x10G 1x1G ] [ ]
port-0/0/18 1x10G - - - 1x10G [ 1x10G 1x1G ] [ ]
port-0/0/19 1x10G - - - 0x0 [ 1x10G 1x1G ] [ ]
port-0/0/20 0x0 - - - 1x10G [ 1x10G ] [ ]
usr_config :
   pic_usr_config :
      pic_name : pic-0/0
      pic_mode_valid : false
      pic_mode :
      num_ports_valid : false
      num_ports : 0
      config_invalid : false
      ptp_mode : true

regress@rtme-acx7020-01:pfe>
```

1.13 Supported Optics

A list of all supported optical modules is available in the following site.

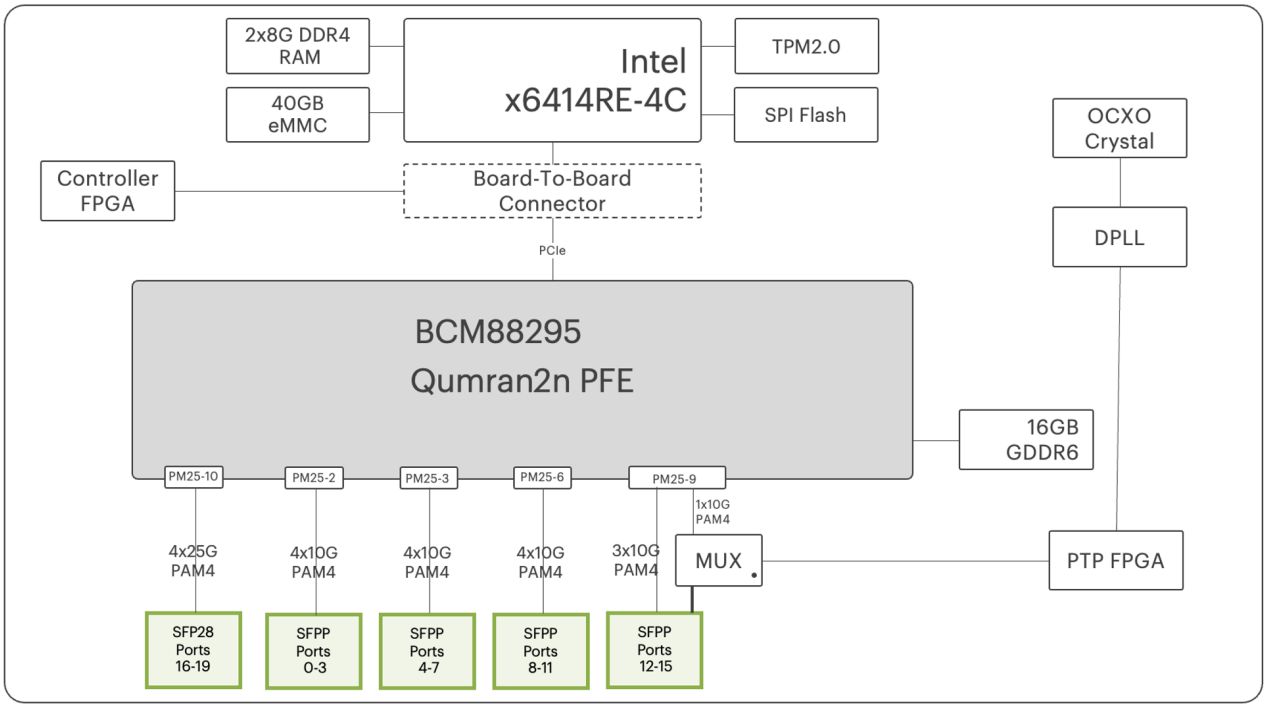
https://apps.juniper.net/hct/product/ACX7020/hwspecs

1.14 Internal Description: Block Diagrams

ACX7020 is internally composed of multiple parts. We can summarize them as:

- A Broadcom Qumran 2n PFE (BCM88295) with 16GB of GDDR6
- An Intel x6414 CPU (4 cores) with 16GB DDR4 memory and SSD storage
- An internal controllers for I2C (inter-integrated circuit), fan control, …
- A PTP FPGA
- A TPM 2.0 module (Trusted Platform Module)

The ACX7020 is a “System on the Chip” (SoC). That means it is based on a single PFE where all parts involved in the forwarding tasks are directly connected. This Qumran PFE is a deep-buffer ASIC with hybrid packet memory (internal and external). The external buffer is present on the side of the package: 16GB of GDDR6 memory.



In this diagram, we are presenting the SerDes at 25G (instead of 25.7875Gbps) and 10G (instead of 10.3125Gbps) for sake of simplicity.

We notice the hardware shared for port 19 and PTP FPGA via mux section. At a single point in time, only one can be used.

We invite you to check the articles to learn more about the ASIC selection process and many more details on the packet forwarding ASIC.

- Building the ACX7000 Series: the PFE: https://techpost.hpe.com/articles/building-the-acx7000-series-the-pfe/article

- Let's Talk About VOQ and DNX Pipeline: https://techpost.hpe.com/articles/voq-and-dnx-pipeline/article

The different components can be monitored via CLI.

```
regress@rtme-acx7020-01:pfe> show system info

                                    DATE + TIME

    Date : Thu Mar 05 08:54:34 2026
    System uptime : 05:33:01
    PFE uptime : 05:32:10


                                       CPU

    Model name : Intel Atom(R) x6414RE Processor @ 1.50GHz
    Nodes : 1
    Cores : 4
    Hyper threads : 4
    Pairs : 0 1 2 3


                                     MEMORY

    Total : 14.76GB
    Free : 6.97GB
    Shared : 209.35MB


                                   PROCESSES

  ------------------------------------------------------------------------------------------------------------------------
      PID NAME MEMORY %CPU UPTIME THREADS PG-FLT
                                                          ________________________________
                                                           TOTAL 80-100 60-80 40-60
  ------------------------------------------------------------------------------------------------------------------------
       7128 aft-sysinfo 82.89MB 2.4 05:32:08 6 0 0 0 34032
       7856 evoaft-jvisiond 115.75MB 0.0 05:32:01 7 0 0 0 37186
       6979 imgd 5.20MB 0.0 05:32:10 2 0 0 0 20412
       7435 mgd 79.41MB 0.0 05:32:06 1 0 0 0 28159
       9234 mgd 17.13MB 0.0 05:31:42 1 0 0 0 8755
      10818 mgd 83.00MB 0.0 05:31:18 1 0 0 0 7079
      19842 mgd 14.39MB 0.0 02:12:39 1 0 0 0 523
       9222 mgd-api 73.82MB 0.0 05:31:42 4 0 0 0 33580
       7932 packetio 259.77MB 0.0 05:32:01 39 0 0 0 23783
       9886 packetio-dds-ag 73.27MB 0.0 05:31:35 6 0 0 0 33458
```

1.15 No-FRU System and Redundancy



In ACX7020 systems, aside from optics, there is no field replaceable Units, and the redundant parts are the power supply modules and the fans.

1.16 Routing Engine

ACX7020 have a single routing engine. That means we don't support any kind of High-Availability (HA) option at the RE level. Also, In-Service System Upgrade (ISSU) is not supported, nor planned.

1.17 Fans

The chassis is equipped with 3 fixed side-to-side fans. They can NOT be replaced, however the cooling redundancy is 2+1. So, the system will stay operational with one fan failure case. A part failing eventually implies a router replacement to restore the 2+1 redundancy.

We verify the fan operation with:

```
regress@rtme-acx7020-01> show chassis fan
     Item Status % RPM Measurement
     Fan Tray 0 Fan 1 OK 33% 8000 RPM
     Fan Tray 0 Fan 2 OK 32% 7900 RPM
     Fan Tray 0 Fan 3 OK 32% 7900 RPM

regress@rtme-acx7020-01>
```

Also, we can check the temperature of various components and also the fan operating speeds with:

```
regress@rtme-acx7020-01> show chassis environment
Class Item Status Measurement
Temp FPC 0 Sensor Q2N Max Reading OK 42 degrees C / 107 degrees F
     Routing Engine 0 CPU Temperature OK 44 degrees C / 111 degrees F
     Routing Engine 0 RE temperature OK 36 degrees C / 96 degrees F
     Routing Engine 0 FPC temperature OK 36 degrees C / 96 degrees F
     Routing Engine 0 Outlet temperature OK 36 degrees C / 96 degrees F
     Routing Engine 0 Inlet temperature OK 27 degrees C / 80 degrees F
     Routing Engine 0 Q2N thermal Diode OK 31 degrees C / 87 degrees F
     Routing Engine 0 DIMM0 Temperature OK 33 degrees C / 91 degrees F
     Routing Engine 0 DIMM1 Temperature OK 34 degrees C / 93 degrees F
     Routing Engine 0 Temp Hotswap Controller OK 34 degrees C / 93 degrees F
     Routing Engine 0 MP2940c Temperature OK 38 degrees C / 100 degrees F
Fan Fan Tray 0 Fan 1 OK 8000 RPM
     Fan Tray 0 Fan 2 OK 7900 RPM
     Fan Tray 0 Fan 3 OK 7900 RPM

regress@rtme-acx7020-01>
```

1.18 AC and DC Power

You notice two PSM on both ends of the router, they can be AC or DC with single PSU or dual. With a 1+1 PSM configuration, we have power redundancy: if one fails, the system will raise an alarm but will stay fully operational.

```
regress@rtme-acx7020-01> show chassis power detail
Chassis Power Voltage(V) Power(W)

Total Input Power 49
  PSM 0
    State: Online
    Capacity 220 W (maximum 220 W)
  PSM 1
    State: Online
    Capacity 220 W (maximum 220 W)

System:
  Zone 0:
     Capacity: 440 W (maximum 440 W)
     Actual usage: 50 W
  Total system capacity: 440 W (maximum 440 W)
```

Note: it is possible to deactivate half of the ports and shut down the associated PFE core. This power-saving mode can reduce the consumption for low port density deployments:

https://www.juniper.net/documentation/us/en/software/junos/chassis/topics/task/configuring-power-saving-mode.html

Typically configured ports “unused” (turning down SerDes) can save up to 5W on 50W.

1.19 Timing Support

The ACX7020 supports:

- Synchronous Ethernet with enhanced ESMC
- PTP G.8275.1 / G.8275.1enh / G.8275.2 Profile
- Meets G.8273.2 T-BC Class C Performance
- IEEE 1588 transparent clock along with G.8273.3 T-TC

Note:

- Sync-E and PTP Hybrid not supported on 1G ports
- Port 19 is disabled when you activate PTP

1.20 MDB Profile support

If you are not familiar with the concept of MDB / Hardware profiles, please check this link first:

https://techpost.hpe.com/articles/acx7000-hardware-profiles/article

Two MDB Profiles are supported: lean-edge (default) and cloud-metro. A reload is required to change the profile configuration.

Default mode: Lean edge profile.

```
[vrf:none] regress@rtme-acx7020-01:~# cli-pfe
regress@rtme-acx7020-01:pfe> show evo-pfemand mdb-info
User Config : Profile:lean-edge Kaps:1
Max possible mac limit for this profile:85000

TableName Type Capacity EntrySize
================================================
TCAM 0 0
KAPS1 KAPS (LPM) 1572864 0
KAPS2 KAPS (LPM) 65536 0
ISEM1 Exact Match 65536 30
ISEM2 Exact Match 81920 30
ISEM3 Exact Match 65536 30
INLIF1 Direct Access 65536 60
INLIF2 Direct Access 32768 60
INLIF3 Direct Access 32768 60
IVSI Direct Access 21846 90
LEM Exact Match 393216 30
IOEM1 Exact Match 49152 30
IOEM2 Exact Match 65536 30
MAP Direct Access 0 0
FEC1 Direct Access 32768 150
FEC2 Direct Access 26215 150
FEC3 Direct Access 32768 150
PPMC Exact Match 98304 30
GLEM1 Exact Match 65536 30
GLEM2 Exact Match 65536 30
EEDB1 EEDB 32768 0
EEDB2 EEDB 32768 0
EEDB3 EEDB 32768 0
EEDB4 EEDB 32768 0
EEDB5 EEDB 16384 0
EEDB6 EEDB 16384 0
EEDB7 EEDB 16384 0
EEDB8 EEDB 16384 0
EOEM1 Exact Match 32768 30
EOEM2 Exact Match 49152 30
ESEM Exact Match 65536 30
EVSI Direct Access 32768 30
SEXEM1 Exact Match 65536 30
SEXEM2 Exact Match 0 0
SEXEM3 Exact Match 0 0
LEXEM Exact Match 131072 30
```

```
RMEP_EM Exact Match 65536 30
```

Alterative option: cloud-metro profile

```
[vrf:none] regress@rtme-acx7020-01:~# cli-pfe
regress@rtme-acx7020-01:pfe> show evo-pfemand mdb-info
User Config : Profile:cloud-metro Kaps:1
Max possible mac limit for this profile:270000

TableName Type Capacity EntrySize
================================================
TCAM 0 0
KAPS1 KAPS (LPM) 917504 0
KAPS2 KAPS (LPM) 65536 0
ISEM1 Exact Match 65536 30
ISEM2 Exact Match 49152 30
ISEM3 Exact Match 49152 30
INLIF1 Direct Access 65536 60
INLIF2 Direct Access 65536 60
INLIF3 Direct Access 65536 60
IVSI Direct Access 21846 90
LEM Exact Match 1245184 30
IOEM1 Exact Match 49152 30
IOEM2 Exact Match 49152 30
MAP Direct Access 0 0
FEC1 Direct Access 26215 150
FEC2 Direct Access 13108 150
FEC3 Direct Access 26215 150
PPMC Exact Match 65536 30
GLEM1 Exact Match 65536 30
GLEM2 Exact Match 65536 30
EEDB1 EEDB 32768 0
EEDB2 EEDB 32768 0
EEDB3 EEDB 32768 0
EEDB4 EEDB 32768 0
EEDB5 EEDB 16384 0
EEDB6 EEDB 16384 0
EEDB7 EEDB 0 0
EEDB8 EEDB 0 0
EOEM1 Exact Match 32768 30
EOEM2 Exact Match 32768 30
ESEM Exact Match 49152 30
EVSI Direct Access 16384 30
SEXEM1 Exact Match 65536 30
SEXEM2 Exact Match 0 0
SEXEM3 Exact Match 0 0
LEXEM Exact Match 131072 30
RMEP_EM Exact Match 65536 30
```

1.21 Conclusion

The ACX7020 router is 1-RU, 100Gbps platform, ideally positioned for small cell site and it can be with 1G/10G/25aggregation. In the next articles, we will continue the description of the ACX7000 portfolio and will dig deeper into the specific features of these products.

1.22 Useful links

- ACX7020 Data sheet: https://www.juniper.net/us/en/products/routers/acx-series/acx7000-family-cloud-metro-routers/acx7000-line/acx7020-cloud-metro-routers-datasheet.html
- Building the ACX7100 Series: the PFE: https://techpost.hpe.com/articles/building-the-acx7000-series-the-pfe/article
- Let's talk about VOQ and DNX Pipeline: https://techpost.hpe.com/articles/voq-and-dnx-pipeline/article
- Hardware Guide: https://www.juniper.net/documentation/us/en/hardware/acx7020/index.html
- Hardware Compatibility Tool: https://apps.juniper.net/hct/product/ACX7020/hwspecs

1.23 Glossary

- FRS: First Revenue Ship
- I2C: Inter-Integrated Circuit
- MDB: Modular DataBase
- PFE: Packet Forwarding Engine
- PSM : Power Supply Module
- PTP: Precision Time Protocol
- S2S: Side to Side
- SKU: Stock Keeping Unit
- TPM : Trusted Platform Module
- PSU: Power Supply Unit

1.24 Acknowledgments

Thanks a lot to Nicolas for the document review.



DC – AI – Routing – Switching – Security – Wireless – Automation

© Copyright 2026 Hewlett Packard Enterprise Development LP. The information contained herein is subject to change without notice. The only warranties for Hewlett Packard Enterprise products and services are set forth in the express warranty statements accompanying such products and services. Nothing herein should be construed as constituting an additional warranty.
Hewlett Packard Enterprise shall not be liable for technical or editorial errors or omissions contained herein.
Trademark acknowledgments, if needed. All third-party marks are the property of their respective owners.

HEWLETT PACKARD ENTERPRISE

HPE.com



